Computational Logic

Soundness and Completeness

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Soundness and completeness of a deductive system $\mathcal{D}$

Given a set Σ of premises and a conclusion ϕ:

Soundness If there is a deduction from ϕ with premises Σ, i.e. $\Sigma \vdash_{\mathcal{D}} \phi$, then the conclusion ϕ is a semantic consequence of Σ, i.e. $\Sigma \models \phi$. If $\vdash_{\mathcal{D}} \phi$ then ϕ is a tautology ($\models \phi$) .

Completeness If ϕ is a semantic consequence of Σ, i.e. $\Sigma \models \phi$, then there is a deduction of ϕ with premises Σ, i.e. $\Sigma \vdash_{\mathcal{D}} \phi$. If ϕ is a tautology ($\models \phi$) then $\vdash_{\mathcal{D}} \phi$.

Why study deductive systems?

In the case of propositional logic, truth tables allow us to determine whether a formula ϕ is a tautology or a semantic sequence.
However,

- Obtaining a deduction can be faster than constructing truth tables and with less space consumption.
- It is possible to have an object that represents the deduction and is therefore easier to manipulate it computationally.
- serves to illustrate the notion of deduction, which is essential in logics whose semantics are more complex and truth tables cannot be used.

Soundness of the natural deduction system *DN*

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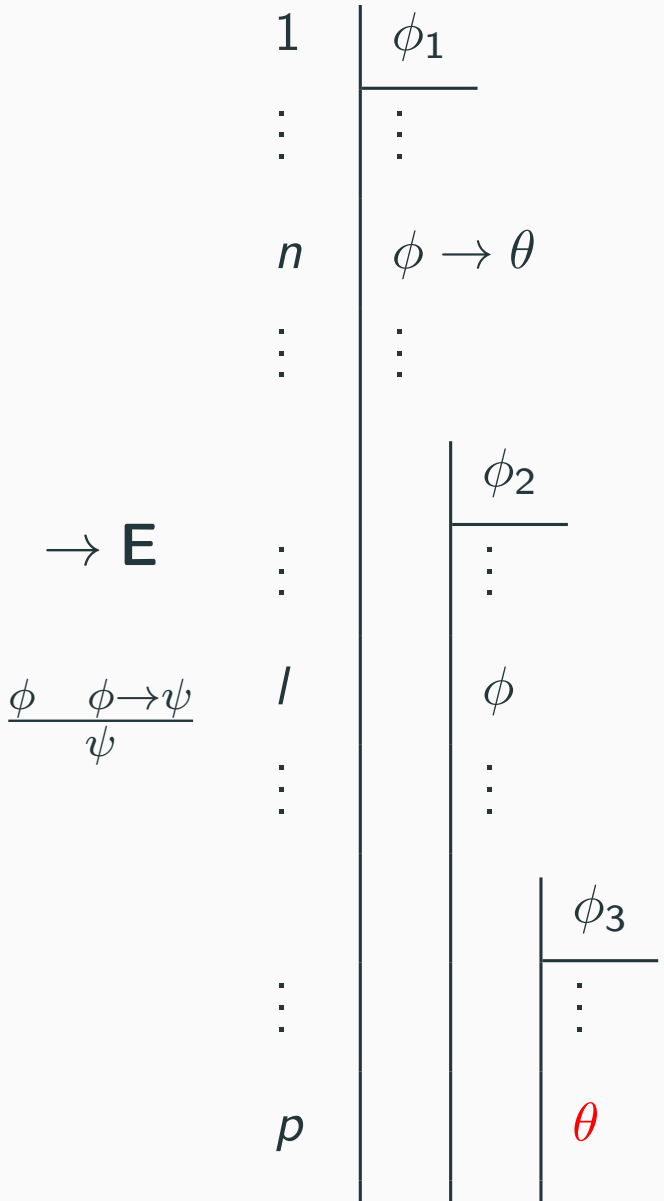
Proposition: If $\Sigma \vdash \phi$ then $\Sigma \models \phi$.

Proof: Suppose we have a deduction of ϕ, $\phi_1, \ldots, \phi_n = \phi$. We will show that at each step the formula that occurs there is a semantic consequence of the premises (or hypotheses) that are assumed there.

We prove by contradiction: Suppose there is a step p that contains a formula that is not a semantic consequence of the premises assumed in p. And let p be the first of these steps. We will see that whatever the *DN* rule applied in p, we have a contradiction. Which allows us to conclude that there is no such step p.

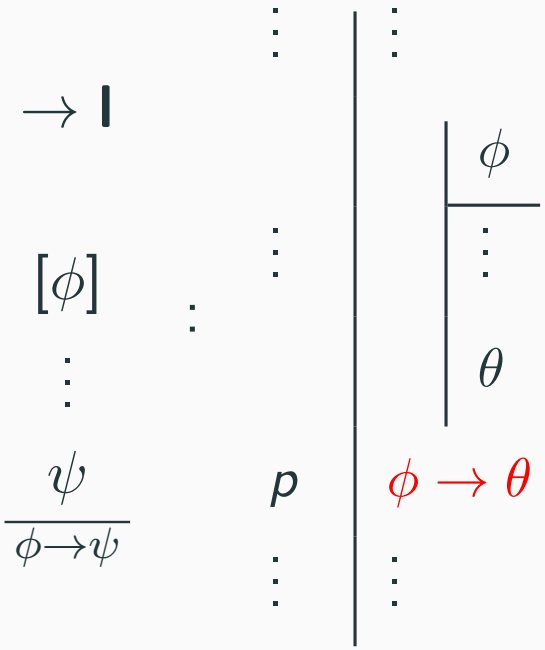
Soundness of the natural deduction system *DN*

We demonstrate this case-by-case, considering each of the rules:



Let θ be the formula deduced in step p by applying $\to$ **E** to $\phi \to \theta$ and ϕ. And let $\phi_1, \ldots, \phi_k$ be the premises assumed in θ, and, by hypothesis, θ is not a semantic consequence of them. But the premises for $\phi \to \theta$ and ϕ are among $\phi_1, \ldots, \phi_k$ and both are semantic consequences of them:

Consider the truth table for the formulas $\phi_1, \ldots, \phi_k, \phi, \phi \to \theta$ and θ. By hypothesis, there exists a valuation v such that $v(\phi_i) = \mathbf{V}$, $1 \leq i \leq k$ and $v(\theta) = \mathbf{F}$. And also $v(\phi) = v(\phi \to \theta) = \mathbf{V}$. But this contradicts the truth table for the implication.



Suppose that step p deduces $\phi \to \theta$ by applying rule $\to$ I to a sub-deduction with premise ϕ and conclusion θ. Let $\phi_1, \ldots, \phi_k$ be the premises assumed in $\phi \to \theta$. In θ the premises are some of $\phi_1, \ldots, \phi_k$ and ϕ and θ is a semantic consequence of them.

Let the truth table be for $\phi_1, \ldots, \phi_k$, ϕ, $\phi \to \theta$ and θ.

Hypothetically, there exists a valuation v such that $v(\phi_i) = \mathbf{V}$, $1 \leq i \leq k$ and $v(\phi \to \theta) = \mathbf{F}$. Then $v(\phi) = \mathbf{V}$ and $v(\theta) = \mathbf{F}$.

But this contradicts the fact that θ is a semantic consequence of $\phi_1, \ldots, \phi_k$ and ϕ.

F E :

$\frac{\mathbf{F}}{\phi}$

Suppose that in step p we deduce ϕ from **F** . Let $\phi_1, \ldots, \phi_k$ be the premises assumed in ϕ, which are the premises assumed in **F** (and of which **F** is a semantic consequence). But this can only be the case if $\phi_1, \ldots, \phi_k$ are not satisfiable. And therefore ϕ is (vacuously) a semantic consequence of $\phi_1, \ldots, \phi_k$.

After analyzing the remaining cases and obtaining a contradiction in all of them, we can conclude that a deduction in the *DN* system cannot have steps that are not a semantic consequence of the premises.

Corollary: If $\vdash \phi$ then $\models \phi$.

Using contraposition we have:

The completeness of natural deduction allows us to determine whether there is not a deduction of a formula ψ from premises $\phi_1, \ldots, \phi_n$ (i.e. $\phi_1, \ldots, \phi_n \nvdash \psi$: it suffices to find a valuation v such that $v(\phi_i) = \mathbf{V}$, $1 \leq i \leq n$ and $v(\psi) = \mathbf{F}$ (i.e. $\phi_1, \ldots, \phi_n \not\models \psi$).

Exercise

Show that $\neg p \ \vee \ (q \rightarrow p) \nvdash \neg p \ \wedge \ q$

Completeness of the deduction system *DN*

Let us see that the natural deduction *DN* is complete for propositional logic: any semantic consequence can be deduced in *DN*; in particular all tautologies are theorems of *DN*.

This seems more complicated than completeness: whenever all assignments satisfy a formula ϕ there is a deduction for it...

Let v be an assignment of values to variables. For each formula ψ, we define

$$\psi^v = \begin{cases} \psi & \text{if } v(\psi) = \mathbf{V} \\ \neg\psi & \text{if } v(\psi) = \mathbf{F} \end{cases}$$

Completeness of the deduction system *DN*

Lemma 1: Let ϕ be a formula whose propositional variables are $q_1, \ldots, q_n$ and let v be an assignment of values to the variables. Then

$$q_1^v, \ldots, q_n^v \vdash \phi^v$$

For example, let $p \wedge q$, $v(p) = \mathbf{V}$ and $v(q) = \mathbf{F}$.

We have that $p^v = p$, $q^v = \neg q$ and $(p \wedge q)^v = \neg(p \wedge q)$.

Then by the lemma, it follows that $p, \neg q \vdash \neg(p \wedge q)$.

Prop 2: If $\models \phi$ then $\vdash \phi$, i.e., if ϕ is a tautology then ϕ is a theorem.

Prop. 3: If $\phi_1, \ldots, \phi_n \models \phi$ then $\phi_1, \ldots, \phi_n \vdash \phi$.

Completeness of the deduction system *DN*

Proof of Prop. 2. We want to show that if $\models \phi$ then $\vdash \phi$.

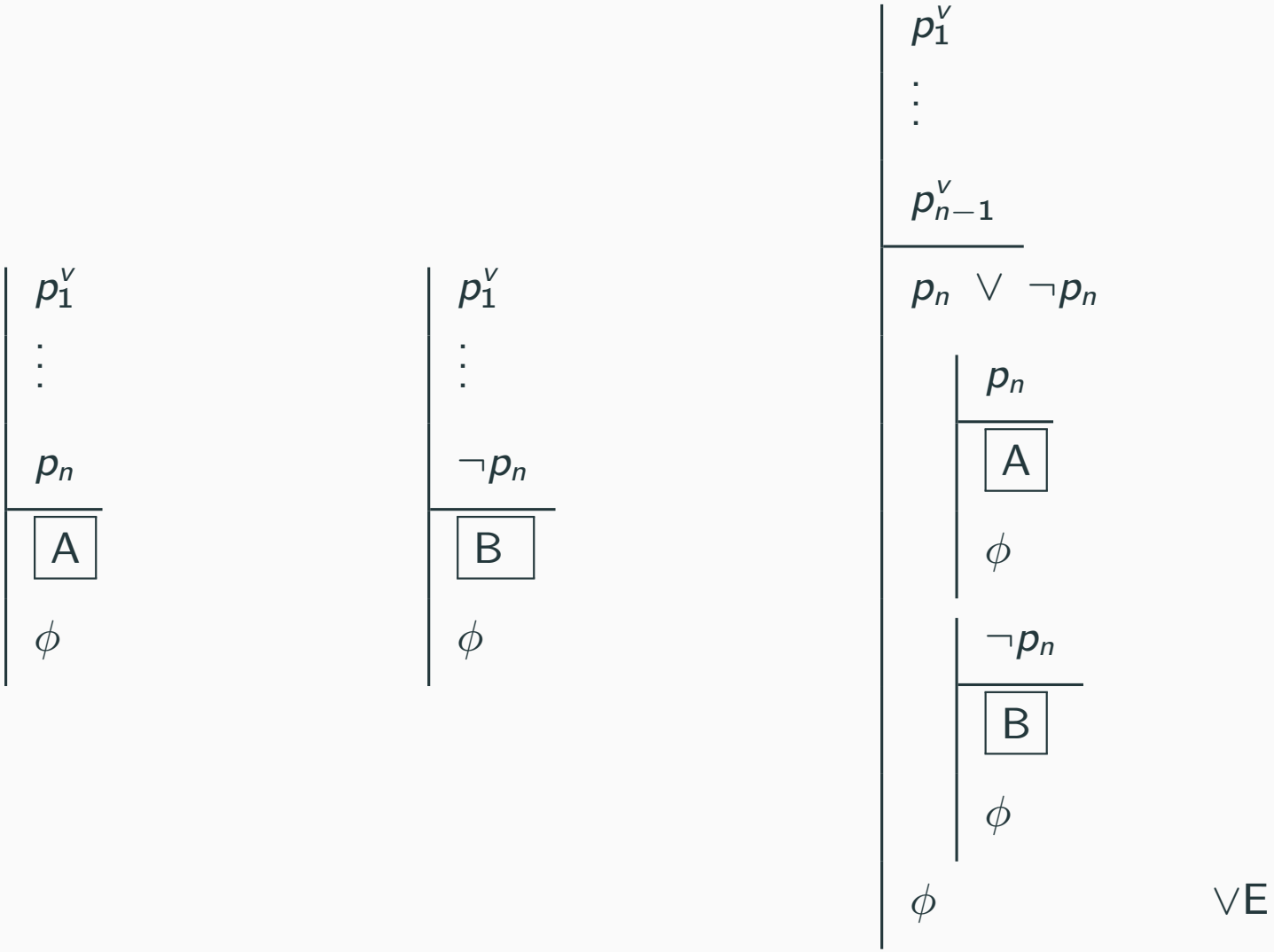
Let ϕ be a tautology and $p_1, \ldots, p_n$ be the propositional variables that occur in it. From $\models \phi$ and by Lemma 1, $v(\phi) = \mathbf{V} = v(\phi^v)$ and $p_1^v, \ldots, p_n^v \vdash \phi$, for every assignment v. We then have::

$$p_1^v, \ldots, p_{n-1}^v, p_n \vdash \phi \text{ and } p_1^v, \ldots, p_{n-1}^v, \neg p_n \vdash \phi$$

for all valuations v. Using TE for $p_n \vee \neg p_n$ and the rule $\vee$ **E**, we can combine the previous deductions into a deduction of:

$$p_1^v, \ldots, p_{n-1}^v \vdash \phi$$

Completeness of the deduction system *DN*



$p_1^v, \ldots, p_{n-1}^v, p_n \vdash \phi$ $p_1^v, \ldots, p_{n-1}^v, \neg p_n \vdash \phi$ $p_1^v, \ldots, p_{n-1}^v \vdash \phi$

Repeating the process $n-1$ times for $p_1, \ldots, p_{n-1}$ we obtain $\vdash \phi$.

Completeness of the deduction system *DN*

Lemma of deduction: $\Sigma \cup \{\phi\} \vdash \psi$ if and only if $\Sigma \vdash \phi \to \psi$.

Proof: We prove the two directions:
$\Rightarrow$: If $\Sigma \cup \{\phi\} \vdash \psi$, we obtain $\Sigma \vdash \phi \to \psi$, assuming ϕ, using the previous derivation until we obtain ψ and applying the rule $\to$ **I**.

$\Leftarrow$: If $\Sigma \vdash \phi \to \psi$, to obtain $\Sigma \cup \{\phi\} \vdash \psi$, we just use deduction of ψ assuming ϕ.

Lemma 4: $\phi_1, \ldots, \phi_n \models \psi$ if and only if
$\models (\phi_1 \to (\phi_2 \to (\ldots(\phi_n \to \psi)\ldots)))$.

Proof: By contradiction. For any valuation
$v((\phi_1 \to (\phi_2 \to (\ldots(\phi_n \to \psi)\ldots)))) = \mathbf{F}$ if $v(\phi_i) = \mathbf{V}$ for all $1 \leq i \leq n$ and $v(\psi) = \mathbf{F}$. But this contradicts $\phi_1, \ldots, \phi_n \models \psi$!

Completeness of the deduction system *DN*

Proof of Prop. 3

By Lemma 4,

$$\models (\phi_1 \to (\phi_2 \to (\ldots(\phi_n \to \psi)\ldots)))$$

By Prop. 2,

$$\vdash (\phi_1 \to (\phi_2 \to (\ldots(\phi_n \to \psi)\ldots)))$$

And by the lemma of deduction,

$$\phi_1, \ldots, \phi_n \vdash \psi$$

The previous result also holds for the case Σ being infinite and therefore combining soundness and completeness, we have for any set of formulas:

$$\Sigma \models \phi \qquad \text{if and only if} \qquad \Sigma \vdash \phi$$

Completeness of the deduction system *DN*

All that is missing is the proof of Lemma 1

Let ϕ be a formula whose propositional variables are $q_1, \ldots, q_n$ and let v be an assignment of values to the variables. Then

$$q_1^v, \ldots, q_n^v \vdash \phi^v$$

Proof; By induction on the structure of the formula ϕ (= the number of connectives occurring in ϕ):

$\phi = q_1$ Then of course $q_1^v \vdash q_1^v$.

$\phi = \neg\phi_1$ and we have $q_1^v, \ldots, q_n^v \vdash \phi_1^v$ by induction hypothesis. If $v(\phi) = \mathbf{V}$, then $v(\phi_1) = \mathbf{F}$, hence $\phi^v = \neg\phi_1 = \phi_1^v$ and therefore $q_1^v, \ldots, q_n^v \vdash \phi^v$. Otherwise, $v(\phi) = \mathbf{F}$, so $v(\phi_1) = \mathbf{V}$, so $\phi_1^v = \phi_1$ and $\phi^v = \neg\neg\phi_1$. We can extend the derivation of $q_1^v, \ldots, q_n^v \vdash \phi_1$ using the rule **DN** and we obtain a derivation $q_1^v, \ldots, q_n^v \vdash \neg\neg\phi_1 = \phi^v$.

$\phi = \phi_1 \circ \phi_2$ where $\circ$ can be $\wedge$, $\vee$ or $\rightarrow$. Let $p_1, \ldots, p_l$ and $r_1, \ldots, r_k$ be, respectively, the propositional variables occurring in ϕ_1 and ϕ_2, and $\{q_1, \ldots, q_n\} = \{p_1, \ldots, p_l\} \cup \{r_1, \ldots, r_k\}$. From $p_1^v, \ldots, p_l^v \vdash \phi_1^v$ and $r_1^v, \ldots, r_k^v \vdash \phi_2^v$ we can deduce, using the rule $\wedge$ **I**:

$$q_1^v, \ldots, q_n^v \vdash \phi_1^v \ \wedge \ \phi_2^v$$

from where we have to deduct ϕ^v.

$\phi = \phi_1 \to \phi_2$ If $v(\phi) = \mathbf{F}$, then $v(\phi_1) = \mathbf{V}$ and $v(\phi_2) = \mathbf{F}$. So $\phi_1^v = \phi_1$ and $\phi_2^v = \neg\phi_2$, and $q_1^v, \ldots, q_n^v \vdash \phi_1 \wedge \neg\phi_2$. To have $q_1^v, \ldots, q_n^v \vdash \neg(\phi_1 \to \phi_2)$, just show that $\phi_1 \wedge \neg\phi_2 \vdash \neg(\phi_1 \to \phi_2)$. If $v(\phi) = \mathbf{V}$, we have 3 cases:

- $v(\phi_1) = v(\phi_2) = \mathbf{F}$. So $\phi_1^v = \neg\phi_1$ and $\phi_2^v = \neg\phi_2$, and $q_1^v, \ldots, q_n^v \vdash \neg\phi_1 \wedge \neg\phi_2$. To have $q_1^v, \ldots, q_n^v \vdash \phi_1 \to \phi_2$, we show $\neg\phi_1 \wedge \neg\phi_2 \vdash \phi_1 \to \phi_2$.
- $v(\phi_1) = \mathbf{F}$ and $v(\phi_2) = \mathbf{V}$. So $\phi_1^v = \neg\phi_1$ and $\phi_2^v = \phi_2$, and $q_1^v, \ldots, q_n^v \vdash \neg\phi_1 \wedge \phi_2$. To have $q_1^v, \ldots, q_n^v \vdash \phi_1 \to \phi_2$, we show $\neg\phi_1 \wedge \phi_2 \vdash \phi_1 \to \phi_2$.
- $v(\phi_1) = v(\phi_2) = \mathbf{V}$. So $\phi_1^v = \phi_1$ and $\phi_2^v = \phi_2$, and $q_1^v, \ldots, q_n^v \vdash \phi_1 \wedge \phi_2$. To have $q_1^v, \ldots, q_n^v \vdash \phi_1 \to \phi_2$, we show $\phi_1 \wedge \phi_2 \vdash \phi_1 \to \phi_2$.

$\phi = \phi_1 \wedge \phi_2$ We have 4 cases:

- $v(\phi_1) = v(\phi_2) = \mathbf{V}$. Then $\phi_1^v = \phi_1$ and $\phi_2^v = \phi_2$, and $q_1^v, \ldots, q_n^v \vdash \phi_1 \wedge \phi_2$ and then $q_1^v, \ldots, q_n^v \vdash \phi_1 \wedge \phi_2 = \phi^v$.
- $v(\phi_1) = \mathbf{F}$ $ev(\phi_2) = \mathbf{V}$. Then $\phi_1^v = \neg\phi_1$ and $\phi_2^v = \phi_2$, and $q_1^v, \ldots, q_n^v \vdash \neg\phi_1 \wedge \phi_2$. For $q_1^v, \ldots, q_n^v \vdash \neg(\phi_1 \wedge \phi_2) = \phi^v$ it is enough that $\neg\phi_1 \wedge \phi_2 \vdash \neg(\phi_1 \wedge \phi_2)$.
- $v(\phi_1) = \mathbf{V}$ $ev(\phi_2) = \mathbf{F}$. Then $\phi_1^v = \phi_1$ and $\phi_2^v = \neg\phi_2$, and $q_1^v, \ldots, q_n^v \vdash \phi_1 \wedge \neg\phi_2$. So that $q_1^v, \ldots, q_n^v \vdash \neg(\phi_1 \wedge \phi_2) = \phi^v$ we need $\phi_1 \wedge \neg\phi_2 \vdash \neg(\phi_1 \wedge \phi_2)$.
- $v(\phi_1) = V(\phi_2) = \mathbf{F}$. Then $\phi_1^v = \neg\phi_1$ and $\phi_2^v = \neg\phi_2$, and $q_1^v, \ldots, q_n^v \vdash \neg\phi_1 \wedge \neg\phi_2$. For $q_1^v, \ldots, q_n^v \vdash \neg(\phi_1 \wedge \phi_2) = \phi^v$ we need $\neg\phi_1 \wedge \neg\phi_2 \vdash \neg(\phi_1 \wedge \phi_2)$.

$\phi = \phi_1 \vee \phi_2$ We have 4 cases:

- $v(\phi_1) = v(\phi_2) = \mathbf{F}$. Then $\phi_1^v = \neg\phi_1$ and $\phi_2^v = \neg\phi_2$, and $q_1^v, \ldots, q_n^v \vdash \neg\phi_1 \wedge \neg\phi_2$. For $q_1^v, \ldots, q_n^v \vdash \neg(\phi_1 \vee \phi_2) = \phi^v$ it is enough that $\neg\phi_1 \wedge \neg\phi_2 \vdash \neg(\phi_1 \vee \phi_2)$.
- $v(\phi_1) = v(\phi_2) = \mathbf{V}$. Then $\phi_1^v = \phi_1$ and $\phi_2^v = \phi_2$, and $q_1^v, \ldots, q_n^v \vdash \phi_1 \wedge \phi_2$. So that $q_1^v, \ldots, q_n^v \vdash \phi_1 \vee \phi_2 = \phi^v$ we need $\phi_1 \wedge \phi_2 \vdash \phi_1 \vee \phi_2$
- $v(\phi_1) = \mathbf{F}$ and $v(\phi_2) = \mathbf{V}$. Then $\phi_1^v = \neg\phi_1$ and $\phi_2^v = \phi_2$, and $q_1^v, \ldots, q_n^v \vdash \neg\phi_1 \wedge \phi_2$. For $q_1^v, \ldots, q_n^v \vdash \phi_1 \vee \phi_2 = \phi^v$ it is enough that $\neg\phi_1 \wedge \phi_2 \vdash \phi_1 \vee \phi_2$.
- $v(\phi_1) = \mathbf{V}$ and $v(\phi_2) = \mathbf{F}$. Then $\phi_1^v = \phi_1$ and $\phi_2^v = \neg\phi_2$, and $q_1^v, \ldots, q_n^v \vdash \phi_1 \wedge \neg\phi_2$. For $q_1^v, \ldots, q_n^v \vdash \phi_1 \vee \phi_2 = \phi^v$ it suffices that $\phi_1 \wedge \neg\phi_2 \vdash \phi_1 \vee \phi_2$.

Decidability of Propositional Logic

Decidability of Propositional Logic

Proposition: Given a set of formulas Σ and a formula ϕ is decidable if $\Sigma \vdash \phi$, i.e., there exists an algorithm that determines whether ϕ is deducible from Σ.

Proposition: By completeness and completeness, deciding $\Sigma \vdash \phi$ is equivalent to deciding whether $\Sigma \models \phi$. And we can suppose $\Sigma = \emptyset$ (using the deduction lemma). Then we just need to construct the truth table for ϕ (which is finite) and check whether ϕ is a tautology.

Decidability of Propositional Logic

Corollary: It is decidable whether a formula ϕ is a theorem (= is valid).

Corollary: It is decidable whether a formula ϕ is satisfiable.

Proof: The formula ϕ is satisfiable if and only if $\neg\phi$ is not a tautology.